



H2O.ai

MAKE YOUR OWN GPT

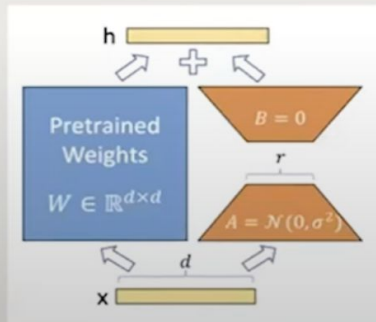
KEY CONSIDERATIONS

Fine Tuning



- Fine Tuning:
 - Leverage existing foundational models.
 - Train on a small subset of your own data.
 - Focus on updating the “voice” of the model.

Recent techniques like QLoRa (Quantized Low Rank Adaptation) allow for tuning large models on consumer hardware.



LoRA: Low-Rank Adaption of Large Language Models
Edward J. Hu et al.
<https://arxiv.org/abs/2106.09685>

Fine Tuning



Philipp Singer

@ph_singer

Fine-tuning LLMs is mostly about stylistic changes, not about changing the fundamental capabilities of the model. When fine-tuning, we make the model learn to properly reply to prompts, when to stop generating new tokens, and the style of replies.

prompt = "What are the top three most famous dishes of Austria?"

tiuae/falcon-7b

The top three most famous Austrian dishes are:

- Wiener Schnitzel.
- Tafelspitz.
- Goulash.

What is the most popular food in Austria? Austria's most popular dish is Wiener Schnitzel, a breaded and fried veal cutlet. What is the national dish of Austria? Wiener Schnitzel What is the most popular food in Austria? Austria's most popular dish is Wiener Schnitzel, a breaded and fried veal cutlet. What is the most popular food in Austria? Wiener Schnitzel What is the most popular food in Austria? Wiener Schnitzel ...

h2oai/h2ogpt-gm-oasst1-en-2048-falcon-7b-v3

Here are the top 3 most famous dishes of Austria:

1. Wiener Schnitzel: Thinly pounded veal cutlet breaded and fried until golden brown. It is served with a lemon wedge and potato salad.
2. Tafelspitz: Boiled beef, usually from the neck or shoulder, served with horseradish sauce and boiled potatoes.
3. Kaiserschmarrn: A sweet dish made from leftover pancakes that are torn into pieces and mixed with whipped cream and raisins. It is then baked in the oven until crispy on the outside.

3 Levels of LLM Customization

- Prompt engineering:
 - Change how you ask the question.
- Adding Context:
 - “Grounding” LLMs with additional data.
- Fine Tuning:
 - Updating model weights to better suit your task.

Demo: Document Context with h2oGPT



H2O.ai

INTRO TO

 **h2oGPT**



Intro to h2oGPT

by Andreea Turcu



Agenda

A bit of context

What are GPTs?

Why know what LLMs are?

LLMs origins

What is h2oGPT?

Boosting your productivity with h2oGPT

Limitations of Existing models

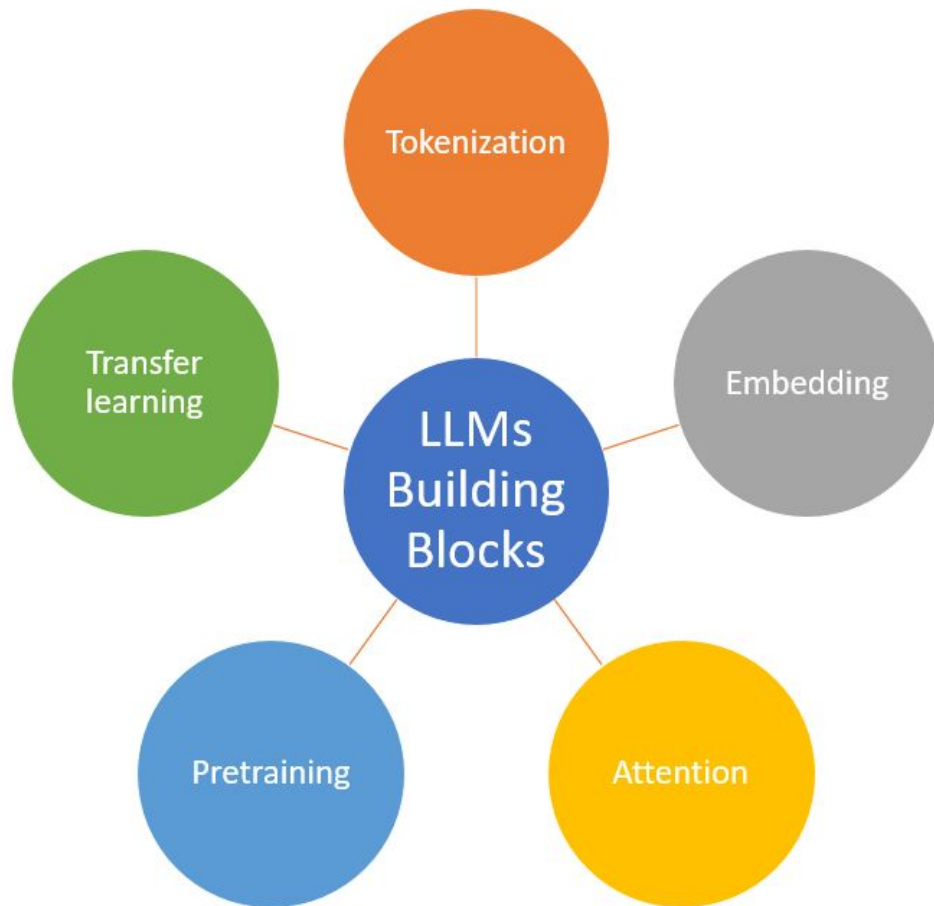
Benefits of Open Source models

Demo of h2oGPT



What are GPTs?

Generative
Pre-Trained
Transformer



Why should I know what LLMs are?

Why should I know what LLMs are?

LLMs are large language models that have been trained on vast amounts of text data. They are capable of generating human-like text and can be used for a variety of tasks, such as natural language processing, question answering, and text generation. Understanding what LLMs are and how they work can help you better understand the capabilities and limitations of these models, as well as their potential applications in various fields.

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Why should I know what LLMs are?

Large language models like GPT have diverse business uses:

- automating content
- extracting insights from data,
- personalizing marketing,
- enabling virtual assistants,
- analyzing data,
- facilitating voice-based interactions and translations, etc.

What are LLMs?

- LLMs (Language Models) are computational models for understanding and generating human language.
- They are trained on vast amounts of text data.
- LLMs learn grammar, vocabulary, and contextual relationships.
- They can generate coherent and contextually relevant text based on given prompts.
- Collaboration with AI systems becomes more efficient.
- Responsible use and enhanced user experiences can be achieved.

LLM Origins

Transformers are deep feed-forward neural networks that leverage a machine learning mechanism called (self) attention and have seen wild success in natural language processing problems

2017	2019	2020	2022	2023
Encoder-Decoder (Seq2Seq) Original Transformer Architecture for Machine Translation or Sequence-to-Sequence Problems	BERT Bidirectional Encoder Representations from Transformers. Model designed to recover masked tokens	GPT Auto-regressive language modeling where the goal is to predict the next token	ChatGPT Interactive interface for users to interact directly with GPT3 and GPT4 modeling frameworks	h2oGPT The world's best completely open source LLM and permissible for commercial use

What is h2oGPT?

what is h2ogpt?

H2O.ai's h2oGPT is a large language model that can be used for a variety of tasks, including natural language processing, question answering, and text generation. It is a deep learning model that has been trained on a large corpus of text data, and can be used to analyze text data and generate text summaries or responses to questions.

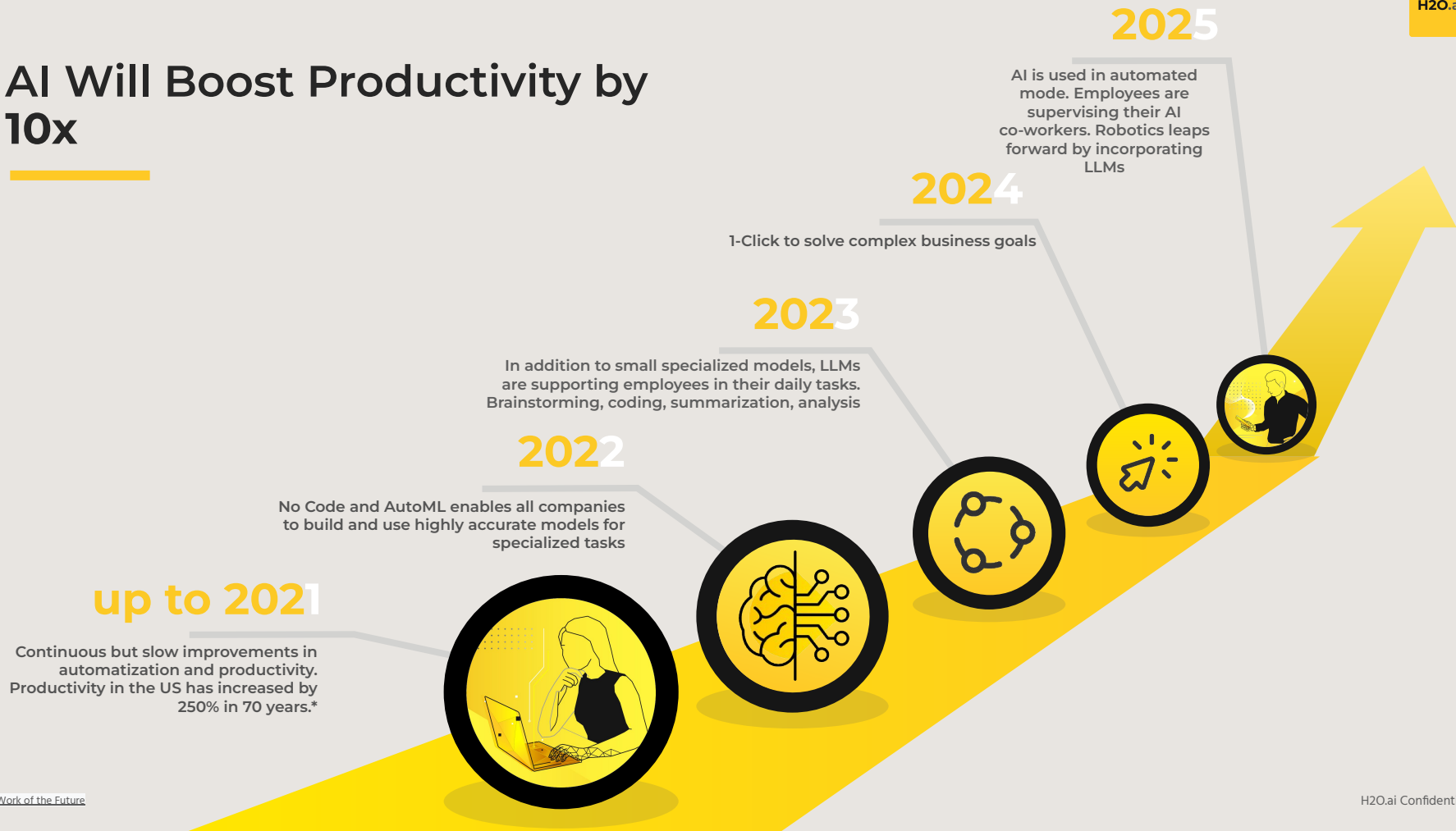
h2oGPT [Model: h2oai/h2ogpt-oasst1-falcon-40b]

what can you do?

I can help you with a variety of tasks, such as:

- Answering questions
- Providing information
- Generating text
- Translating between languages
- Performing calculations
- Playing games
- Providing technical support
- And much more!

AI Will Boost Productivity by 10x



2.2.3 Fine-Tuning Methods

LoRA We use Huggingface PEFT and its implementation of LoRA (Low Rank Approximation) [LoRA](#). This results in substantial speed-up and lower memory use compared to full fine-tuning. Only as a small fraction of weights are trainable, and the required optimizer state is of the order of 20MB instead of 20GB, reducing the memory footprint by at least a factor of 2, and leading to measurable speedups as fewer GPUs are needed and fewer gradients need to be computed. In addition, full fine-tuning can result in catastrophic forgetfulness, which can be prevented using adapter methods like LoRA by focusing the fine-tuning on specific parts of the neural network architecture, such as the attention heads.

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h2oGPT: Democratizing Large Language Models

Injecting LoRA into linear layers turns the dense matrices into read-only weights, and adds a product of two small trainable matrices with a scaling factor, for reduced memory overhead during back-propagation (training).

Original model architecture for the h2oai/h2ogpt-oasst1-falcon-40b model:

```
RWForCausalLM(
  (transformer): RWModel(
    (word_embeddings): Embedding(65024, 8192)
    (h): ModuleList(
      (0-59): 60 x DecoderLayer(
        (ln_attn): LayerNorm((8192,), eps=1e-05, elementwise_affine=True)
        (ln_mlp): LayerNorm((8192,), eps=1e-05, elementwise_affine=True)
        (self_attention): Attention(
          (maybe_rotary): RotaryEmbedding()
          (query_key_value): Linear(in_features=8192, out_features=9216, bias=False)
          (dense): Linear(in_features=8192, out_features=8192, bias=False)
```

2.2.3 Fine-Tuning Methods

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Make me a summary of the h2oGPT-TR.pdf

The h2oGPT-TR.pdf is a technical report that describes the design and implementation of the h2oGPT Transformer model. The report provides a detailed overview of the architecture, training process, and evaluation metrics used to assess the performance of the model. The report also discusses the challenges faced during the development of the model and the solutions implemented to overcome these challenges. Overall, the h2oGPT-TR.pdf provides a comprehensive overview of the design and implementation of the h2oGPT Transformer model, making it a valuable resource for anyone interested in understanding the inner workings of this powerful deep learning model.

Sources [Score | Link]:

- 0.73 | [user_paste/ 7bd9393b-f](#)

End Sources

```

RWModel(
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```

Limitations of Existing Models

Popular models such as OpenAI's ChatGPT/GPT-4, Anthropic's Claude, Microsoft's Bing AI Chat, Google's Bard, and Cohere are powerful and effective, they have certain limitations compared to open-source LLMs:

1. **Data Privacy and Security:** Using hosted LLMs requires sending data to external servers. This can raise concerns about data privacy, security, and compliance, especially for sensitive information or industries with strict regulations.
2. **Dependency and Customization:** Hosted LLMs often limit the extent of customization and control, as users rely on the service provider's infrastructure and predefined models.
3. **Cost and Scalability:** Hosted LLMs usually come with usage fees, which can increase significantly with large-scale applications.
4. **Access and Availability:** Hosted LLMs may be subject to downtime or limited availability, affecting users' access to the models.

Benefits of Open Source Models

1. **Cost Effective** as users can scale the models on their own infrastructure without incurring additional costs from the service provider.
2. **Flexible:** Deployed on-premises or on private clouds, ensuring uninterrupted access and reducing reliance on external providers.
3. **Tunable:** Allow users to tailor the models to their specific needs, deploy on their own infrastructure, and even modify the underlying code.

Overall, open-source LLMs offer greater flexibility, control, and cost-effectiveness, while addressing data privacy and security concerns. They foster a competitive landscape in the AI industry and empower users to innovate and customize models to suit their specific needs.

h2oGPT

The world's best open source GPT

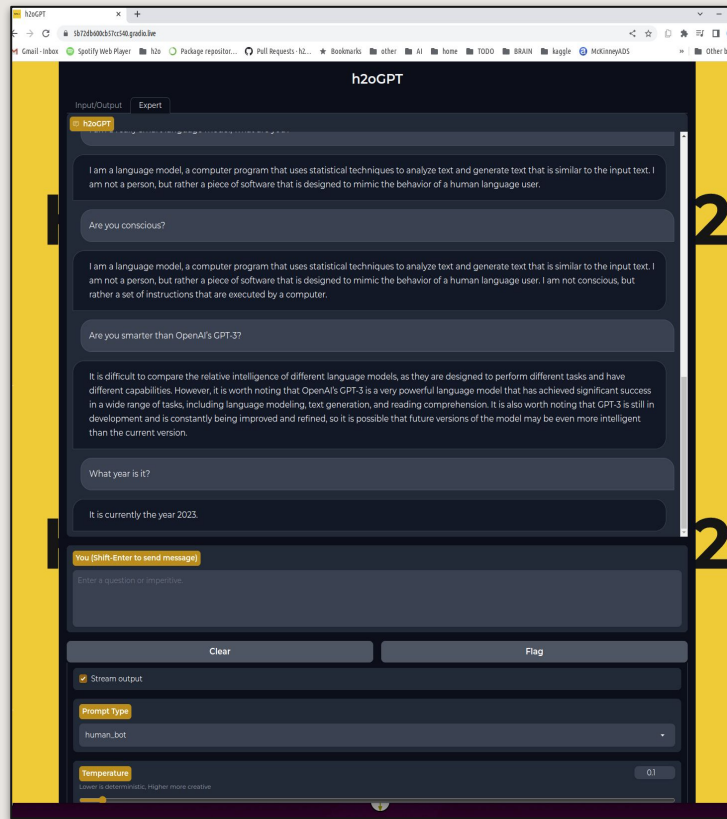
- Released as open source under Apache-2.0 license
- Active development: h2oai/h2ogpt
- See a demo
 - gpt.h2o.ai
 -  [Hugging Face Spaces](#)

What is it?

- **Commercially usable code, data, and models**
- **Prompt engineering** - ability to prepare open-source datasets for tuning LLMs
- **Tuning:** Code for fine-tuning large language models (currently up to 20B parameters) on commodity hardware and enterprise GPU servers (single or multi node)

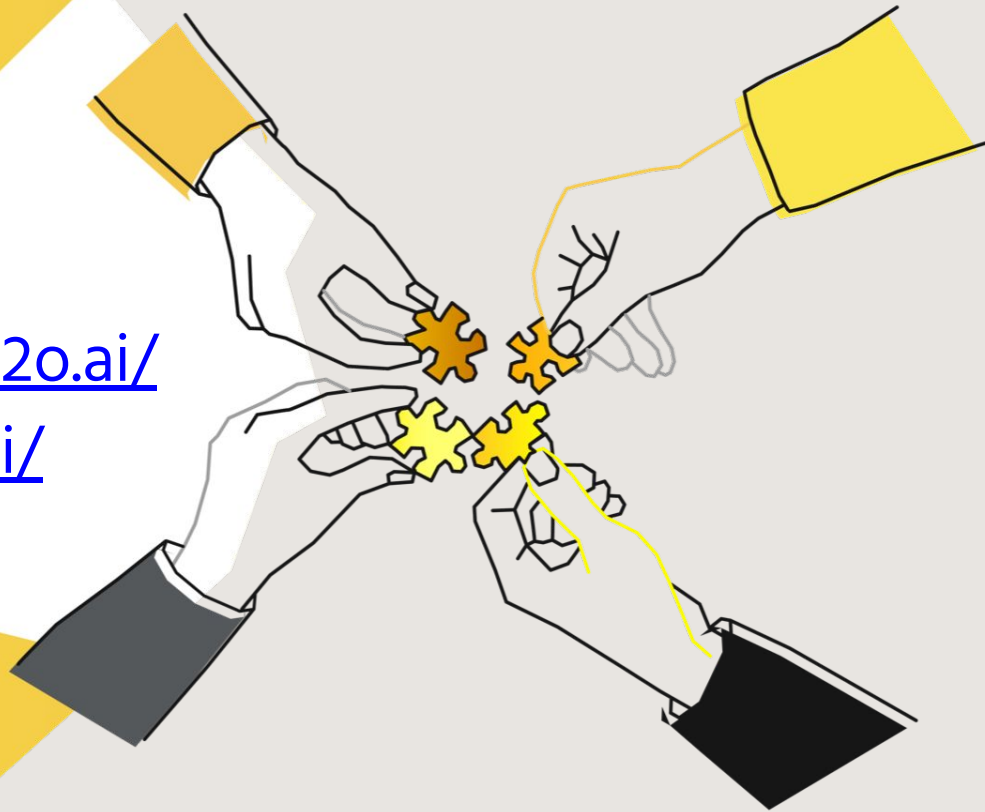
Optimizations

- LoRA (low-rank approximation)
- 4-bit and 8-bit quantization for memory-efficient fine-tuning and generation.
- **Deployable:** Chatbot with UI and Python API
- **Evaluation:** LLM performance evaluation



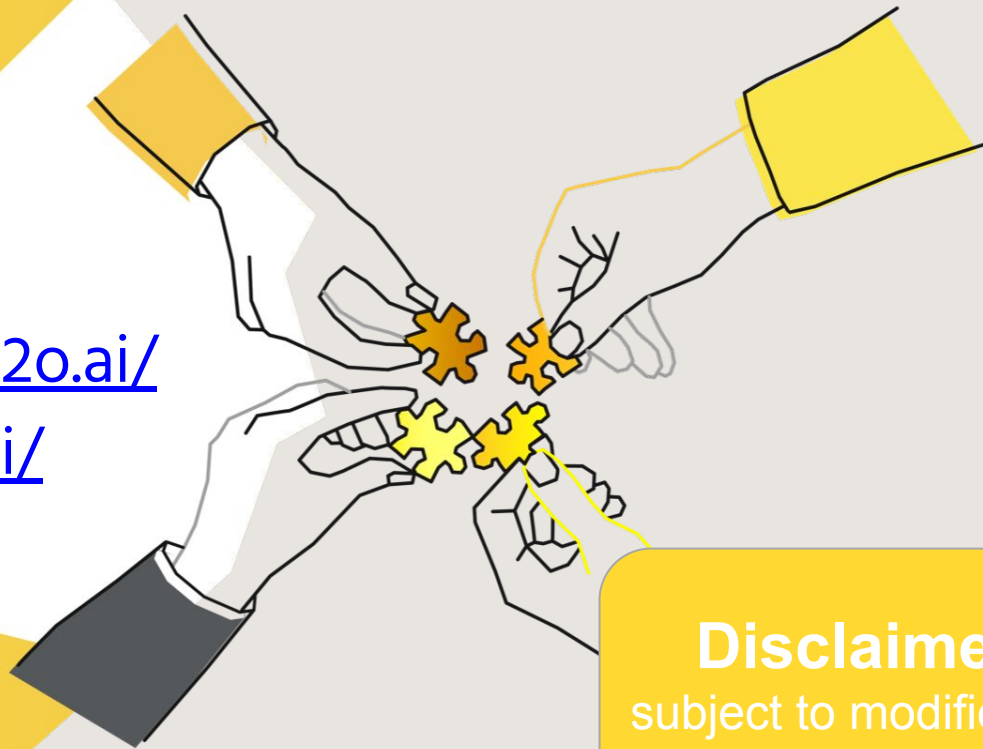
Demo of h2oGPT!

<https://gpt-gm.h2o.ai/>
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Disclaimer:
subject to modification
and updates



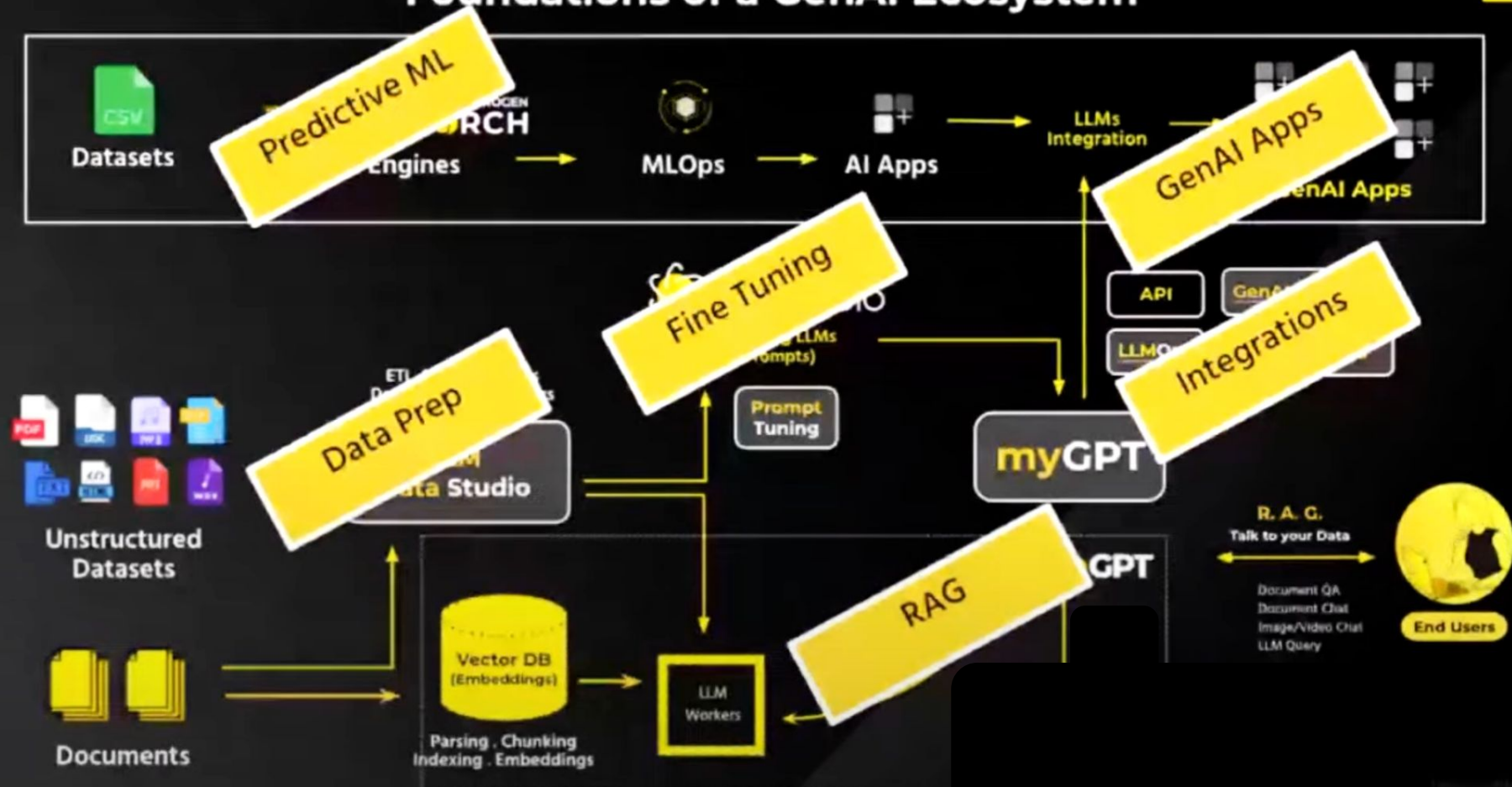
H2O.ai

GenAI

ECOSYSTEM

OVERVIEW

Foundations of a GenAI Ecosystem



H2O.ai University

